WHEN IMAGINATION HURTS: SCORE-TAIL AUDITS FOR WORLD-ACTION ROLLOUT PLANNING

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ABSTRACT

World-action planners often spend test-time compute by imagining candidate futures, scoring those futures, and executing the highest-scoring rollout. This can help only when the planner-score tail contains high real utility; when the tail is exhausted or misaligned, more imagination can saturate or harm. We frame this as a score-tail audit problem for a fixed rollout generator and fixed scorer. For finite empirical rollout pools, we give an exact tie-aware identity for expected selected real utility under i.i.d. with-replacement sampling, max-score selection, and uniform maximum-score tie-breaking. The identity turns the selected-utility curve into an audit of real utility in the upper score tail. Binary success yields an exact $N = 2$ tie-aware AUC identity, while larger candidate counts depend on higher upper-tail rank moments beyond AUC. Experiments freeze a RAM-light protocol over committed artifacts: analytic identities, learned WAM-lite models, bad-scorer falsifications, imagined-real mismatch, adaptive-budgeting runs, and 8 simulated robotics benchmark rows spanning 454 heldout rollout pools. A completed serial LIBERO run adds a 40/40 configured-task CPU rollout-pool audit with 80 state-level pools and positive learned-vs-random lower confidence bound. A V5 prospective layer adds 50 hash-locked heldout audit decisions, a 1458-row finite-pool census, ScoreTailBench core pools, and low-RAM CPU reproduction. A V6 real-benchmark layer freezes cross-family decisions over 543 existing simulated benchmark rollout-pool curves from 10 families, with 99.0% decided accuracy and 0.2% false-allow rate. The paper is not a WAM training scaling law, not real-world robot validation, not modern VLA-scale validation, not full RoboCasa-wide validation, and not a universal OOD/generalization guarantee.

1 INTRODUCTION

Modern embodied agents increasingly act by imagining candidate futures. A learned world model or generative policy proposes rollouts; a planner, verifier, value model, reward model, or hand-designed heuristic scores them; the planner executes the highest-scoring rollout, or the first action of it. Related verifier and best-of-N alignment work studies the same sample-many/rerank primitive (Cobbe et al., 2021; Beirami et al., 2024). This pattern appears in latent world-model planning (Ha & Schmidhuber, 2018; Hafner et al., 2019; 2020; 2023; Hansen et al., 2024), model-based reinforcement learning and sampling MPC (Chua et al., 2018; Rubinstein, 1999; Williams et al., 2017), and the broader move toward generative robot policies and vision-language-action systems (Brohan et al., 2022; 2023; Chi et al., 2023; Octo Model Team et al., 2024).

This makes test-time rollout sampling a basic inference knob. If a planner or simulated agent can sample 1, 8, 32, or 128 candidate futures before acting, how many should it sample? The usual answer is operational rather than conceptual: sample more when compute allows. But sampling more candidates selects deeper into the planner-score tail. If that tail contains truly high-utility rollouts, extra imagination helps. If that tail contains rollouts that only look good under the model or scorer, extra imagination can saturate or even reduce real utility. The central object is therefore not merely the average model error, the average reward prediction, or a pairwise ranking statistic. It is the real utility of rollouts that the planner-score tail actually exposes.

This paper isolates a score-tail audit from the training problem. For a fixed generator and scorer, finite empirical rollout pool, i.i.d. with-replacement sampling, max-score selection, and uniform

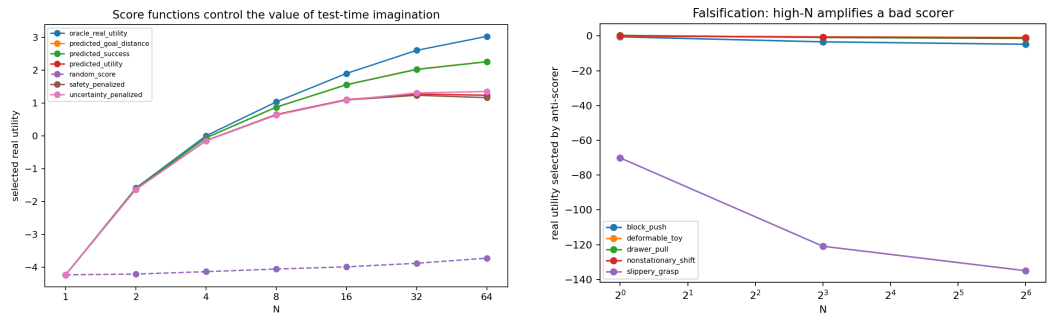

Figure 1: **The value of imagination is a score-tail phenomenon.** Left: aligned learned and oracle-like scores improve selected real utility as N grows, while a random score remains poor. Right: a deliberately anti-aligned scorer can make high-N selection worse. The audit explains both curves with the same object: the joint distribution of planner score and real utility in the upper score tail.

maximum-score tie-breaking, the expected selected real utility is exactly determined by the joint distribution of planner score and real rollout utility. We use that identity as an audit: aligned tails justify larger candidate budgets, exhausted tails suggest stopping, and misaligned tails warn that additional imagined rollouts can amplify the wrong futures.

The binary setting exposes a sharp boundary for familiar ranking summaries. For $N = 2$, selected success is determined by the success rate and a tie-aware AUC term. For $N > 2$, AUC alone is insufficient: larger candidate pools depend on higher upper-tail rank moments. Thus the same two-sample ranking quality can lead to different high-N rollout-selection value.

We check this picture with artifact-backed experiments. Controlled analytic cases test the finite identity, the AUC boundary, and same-AUC/high-N counterexamples. Learned WAM-lite and simulated rollout-pool artifacts illustrate useful, saturated, and harmful regimes. Bad-scorer and imagined-real mismatch experiments directly attack the dangerous case: the model score improves while real utility does not. We also use the resulting inference-value profile as a fixed-stack adaptive budgeting heuristic: estimate whether another rollout is likely to buy real utility before spending the sample.

The scope is deliberately narrow. We do not propose a universal world-action-model training law, a Robot Chinchilla-style training rule, a real-world robot result, a modern VLA-scale validation, a full RoboCasa-wide validation, or a universal OOD/generalization guarantee. Instead, we ask:

> *For a fixed rollout generator and fixed scorer, what is the expected real utility of executing the highest-scoring rollout among N imagined candidates?*

Contributions. First, we define a score-tail audit for fixed world-action rollout generators and scorers: measure real utility in the planner-score tail that max-score selection will actually choose from. Second, we give the exact finite empirical, tie-aware identity that computes selected real utility for any candidate count, real-valued utility, binary success labels, and score ties. Third, we locate the AUC boundary: tie-aware AUC is complete for two candidates, while larger candidate pools depend on higher upper-tail rank moments. Fourth, we turn the identity into an inference-value profile for improvement, saturation, harm, and adaptive rollout budgeting. Fifth, experiments stress-test the audit in analytic, learned WAM-lite, mismatch, falsification, and simulated robotics rollout-pool artifacts under frozen gates with explicit baselines, negative controls, artifact hashes, and scoped claims, including a complete 40-task configured LIBERO serial CPU rollout-pool run. Sixth, the V5 layer adds prospective heldout decisions, equal-compute CPU frontiers, a small closed-loop audit harness, impossibility-boundary examples, and ScoreTailBench packaging. Seventh, the V6 layer transfers the audit across existing real benchmark rollout-pool curve artifacts with calibration, abstention, robustness, leakage hashes, finite-sample accounting, and compute-normalized ablations, without changing the real-robot and SOTA nonclaims.

2 Problem Setup

Fix a planning state x. A rollout generator samples candidate action/future rollouts Z. Each sampled rollout has a planner score $S = s(Z)$ used for selection and a real utility $R = r_{\mathrm{real}}(Z)$ measured under the true environment or heldout simulator. In binary tasks, $R = Y \in \{0, 1\}$.

An N-candidate world-action planner samples N rollouts independently from the fixed rollout distribution, selects a maximum-score sampled rollout, and breaks maximum-score ties uniformly at random. The inference value is

$$V_N(x) = \mathbb{E}[\text{real utility of the selected rollout} \mid x].$$

For binary success, we write $f_N(x) = V_N(x)$.

The fixed-x formulation is intentional. In a receding-horizon controller, the law applies conditionally at each visited state. It does not imply a global closed-form law for arbitrary closed-loop dynamics. Likewise, the theorem assumes that the relevant score/utility distribution or empirical pool is known; estimating it from pilot rollouts is a separate statistical problem.

3 Score-Tail Rollout Audit

The audit is powered by a finite empirical identity. This is the form used for tied empirical rollout pools in our experiments.

Theorem 1 (Finite score-tail rollout identity). *Fix a planning state x and a finite empirical rollout pool $\mathcal{D}_x = \{(S_i, R_i)\}_{i=1}^m$. Sample N rollouts independently and uniformly with replacement from $\mathcal{D}_x$. Select a maximum-score sampled rollout, breaking maximum-score ties uniformly at random. Sort the pool by score in ascending order and group equal-score ties. Tie group g occupies one-indexed ranks $[r_{\min,g}, r_{\max,g}]$ and has mean real utility $\bar{R}_g$. Then*

$$V_N(x) = \sum_g \bar{R}_g \left[\left(\frac{r_{\max,g}}{m} \right)^N - \left(\frac{r_{\min,g} - 1}{m} \right)^N \right].$$

Interpretation. The bracketed term is the probability that the maximum score among the N samples lands in tie group g. The expected selected utility is therefore a weighted average of tie-group real utilities, where higher score groups receive increasing mass as N grows. This is the score-tail audit: the curve improves, saturates, or declines according to the real utility in the score groups that receive selection mass.

Binary success. For $R = Y \in \{0, 1\}$, the same theorem gives the selected success probability $f_N(x)$. Binary success is a special case of the utility-valued law, not a separate mechanism.

Population no-tie form. If scores are continuous and ties occur with probability zero, then

$$V_N(x) = N\,\mathbb{E}\big[R F_S(S)^{N-1} \mid x\big], \tag{1}$$

where F_S is the score CDF at state x. For binary success with prevalence $p = \Pr(Y = 1)$, successful score $S_+ \sim S \mid Y = 1$, and marginal score CDF F_{mix},

$$f_N(x) = Np\,\mathbb{E}\big[F_{\mathrm{mix}}(S_+)^{N-1} \mid x\big]. \tag{2}$$

Equations 1 and 2 are useful population intuitions. The finite theorem remains the correct empirical law when scores tie.

4 AUC for Two Samples, Tail Moments for Many

The binary law connects cleanly to AUC only at $N = 2$, using the standard tie-aware ranking interpretation of ROC AUC (Hanley & McNeil, 1982).

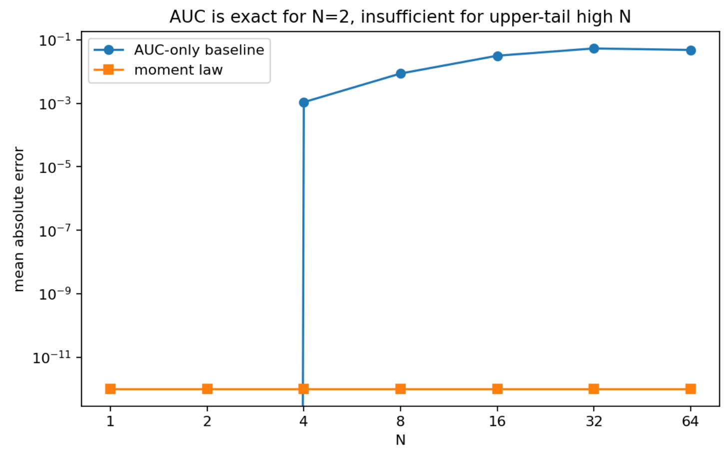

Figure 2: **AUC is exact for $N = 2$ and insufficient for high N.** AUC-only extrapolation matches the two-sample identity but misses upper-tail behavior as N grows. The moment law tracks the full N curve because it uses $\mathbb{E}[U^{N-1}]$.

Proposition 1 ($N = 2$ tie-aware AUC identity)**.** *Let $p = \Pr(Y = 1)$, let S_+ and S_- denote scores conditioned on success and failure, and define tie-aware AUC*

$$\kappa = \Pr(S_+ > S_-) + \frac{1}{2}\Pr(S_+ = S_-).$$

Then

$$f_2 = p^2 + 2p(1-p)\kappa.$$

Interpretation. With two samples, success is guaranteed when both sampled candidates succeed; when exactly one succeeds, the planner succeeds exactly when the successful candidate wins the score comparison, with half credit for ties.

For $N > 2$, AUC is not enough. Define the successful rollout's marginal score rank

$$U = F_{\text{mix}}(S_+).$$

Equation 2 becomes

$$f_N = Np\,\mathbb{E}[U^{N-1}].$$

AUC fixes the first rank moment needed for $N = 2$, but larger N depends on higher powers of U. These higher moments emphasize the upper score tail. Two scorers can therefore have the same p and the same tie-aware AUC while producing very different high-N value.

5 Inference-Value Profiles as Audits

The identity turns $N \mapsto V_N(x)$ into an *inference-value profile* for a fixed planning state, generator, scorer, and real utility distribution. The profile is a score-tail accounting object. If high-score rollouts also have high real utility, the curve improves with more samples. If the useful high-score tail is exhausted, the curve saturates. If high-score rollouts have low real utility, the curve can decline.

This view separates score optimization from deployment value. World-action planning is useful because imagined futures can expose good actions before acting. It is risky because the planner selects by score, not by real utility. This echoes model-bias and model-exploitation concerns in model-based reinforcement learning (Janner et al., 2019). If a score aligns with imagined utility but the high-score tail is misaligned with real outcomes, increasing N over-selects rollouts that look good under the model while failing under the environment.

The identity does not fail in this setting. It predicts selected real utility from the joint distribution of score and real utility. The failure is score/utility alignment. This gives score-tail accounting for how high-N selection can amplify mismatch in tested settings.

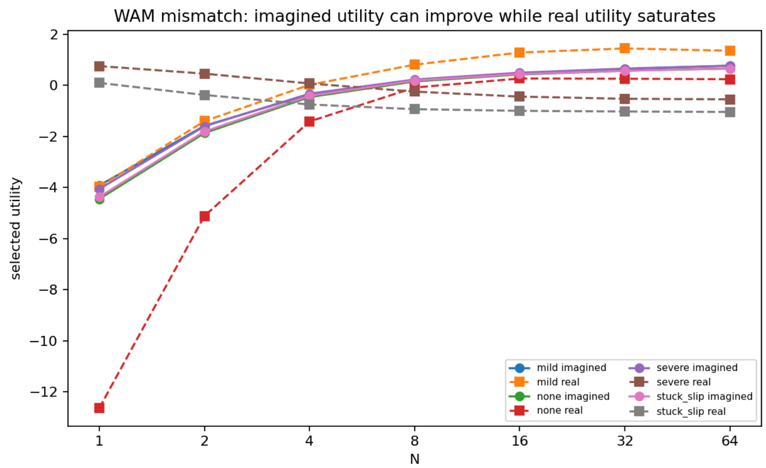

Figure 3: **Imagined-real mismatch can hide behind improving model scores.** As N increases, selected imagined utility can improve while selected real utility saturates or degrades. The law says to audit real utility in the high-score tail, not just imagined value.

6 EXPERIMENTS

Each experiment family checks an audit claim or illustrates a world-action planning failure mode.

Finite identity and AUC boundary. Finite empirical checks compare the tie-aware identity with Monte Carlo selection. The claim ledger reports binary success MAE 0.00210, utility MAE 0.01376, and $N = 2$ AUC identity max error 0.0 in core checks. These tests check the implementation of Theorem 1 and Proposition 1.

AUC versus upper-tail moments. Controlled same-p, same-κ examples show that AUC-only summaries can miss high-N behavior. The moment-law predictor remains accurate because it uses the higher rank moments in $f_N = Np\,\mathbb{E}[U^{N-1}]$ (Figure 2).

Score alignment regimes. Scorer comparisons include random, learned, heuristic, oracle, safety-penalized, uncertainty-penalized, and deliberately bad scorers. Figure 1 shows both aligned and anti-aligned behavior. The empirical pattern matches the audit: useful imagination requires real utility in the upper score tail.

Bad scorer and imagined-real mismatch. Falsification tests use anti-aligned scorers and compare selected imagined utility with selected real utility (Figure 3). These checks illustrate the risk that high-N selection concentrates probability mass on low-real-utility rollouts in the high planner-score tail.

Learned WAM-lite checks. Learned WAM-lite artifacts check whether the same inference-value accounting appears beyond analytic score functions. These experiments report learned model errors, finite-identity agreement on heldout rollout pools, and learned-scorer versus random-scorer comparisons. They are lightweight learned-model checks, not evidence for a universal WAM training recipe.

Simulated robotics rollout pools. We also evaluate in learned WAM-lite and simulator/benchmark rollout-pool artifacts. Verified finite-identity MAEs include Gymnasium Robotics Fetch 0.0126, Gymnasium Robotics RGB visual WAM 0.0106, ManiSkill state-mode 0.00345, Meta-World 0.0298, RoboSuite 0.00237, RoboCasa 12-task family 0.000292, RoboCasa residual 35-task clean/cook 0.000246, and full-suite serial LIBERO 0.000148. Learned scorers beat random with positive confidence intervals in several rollout-pool benchmarks, including Fetch, Meta-World, RoboSuite,

LIBERO slice	Tasks	Pools	Mean Δ_8	95% CI lower
All configured	40	80	0.371	0.318
Spatial	10	20	0.317	0.203
Object	10	20	0.340	0.249
Goal	10	20	0.484	0.397
LIBERO-10	10	20	0.343	0.228

Table 1: **Main-paper LIBERO full-suite signal.** The completed serial run covers 40/40 configured LIBERO tasks and 80 state-level rollout pools. Δ_8 is the promoted learned scorer minus random selected real utility at $N = 8$; the last column reports the suite-level confidence lower bound. Task-level diagnostics, including worst-task means, are retained in the generated artifacts. This is CPU rollout-pool evidence, not real-robot validation, not VLA-scale SOTA, and not solved-policy success.

Scorer	$N = 32$ utility	Δ vs. random	Role
Random scores	0.411	–	Baseline
Zero-shot text	0.443	$+0.033$ [-0.189, 0.254]	Not significant
CLIP + ridge	0.556	$+0.146$ [0.010, 0.281]	Positive frozen-feature tail
Shuffled ridge	0.472	$+0.061$ [-0.046, 0.168]	Negative control
Low-energy	0.891	$+0.480$ [0.358, 0.602]	Strong simple baseline
Oracle	0.984	$+0.573$ [0.469, 0.677]	Ceiling

Table 2: **Frozen visual inference stress test.** A locked CLIP ViT-B/32 probe scores 576 heldout RGB final-frame candidates from 18 Fetch rollout pools. Utilities are pool-normalized; intervals are 95% CIs over task-level seeds for $N = 32$. The result is deliberately not a VLA-scale claim: zero-shot text is unreliable, a tiny ridge head on frozen CLIP embeddings gives a modest positive tail, and the hand-designed low-energy heuristic remains much stronger.

RoboCasa families, and LIBERO rollout-pool checks. Table 1 foregrounds the completed LIBERO run: all 40 configured tasks across Spatial, Object, Goal, and LIBERO-10 suites are present, the aggregate $N = 8$ learned-vs-random lower confidence bound is positive, and every suite-level lower bound remains positive. Task-level diagnostics are retained in the generated artifacts, but the main claim is suite-level rollout-pool evidence rather than per-task dominance. These are simulated CPU rollout-pool results, not hardware evidence, not solved-policy success, and not full modern VLA policy validation.

Frozen visual inference stress test. Recent inference-time sampling work shows that frozen base models can sometimes recover capabilities through sampling alone (Karan & Du, 2026). To connect that philosophy to rollout selection without claiming a new robotics post-training recipe, we freeze CLIP ViT-B/32 and score existing RGB rollout-final frames. Table 2 shows the guarded result: zero-shot language scoring is not significantly above random at high N, frozen CLIP embeddings plus a tiny ridge calibrator give a modest positive tail, and the low-energy heuristic remains far stronger. The audit gain over blind $N = 32$ is positive for zero-shot text ($+0.138$ [0.080, 0.196]) and CLIP+ridge ($+0.099$ [0.039, 0.159]), so the value is diagnostic and budget-controlling rather than a SOTA visual policy claim.

7 V4 Evidence Audit

The v4 paper adds a frozen evidence layer over the repository artifacts. The script `experiments/v4_frozen_evidence.py` reads the committed theorem, mismatch, allocation, robotics, RoboCasa, LIBERO, claim-ledger, and artifact manifest summaries. It writes compact CSV ledgers, six figures, a summary JSON file, and this manuscript's macro file. The layer is intentionally light: it does not rerun optional RoboCasa or LIBERO environments, and it does not promote unavailable hardware or VLA-scale claims.

The current repository-wide ledger reports 150 verified claim checks, 0 partial claim checks, and 0 unsupported claim checks under the repository's claim-status script. The artifact manifest covers 462 files, including 232 CSV tables and 149 JSON summaries. These counts are not a claim of

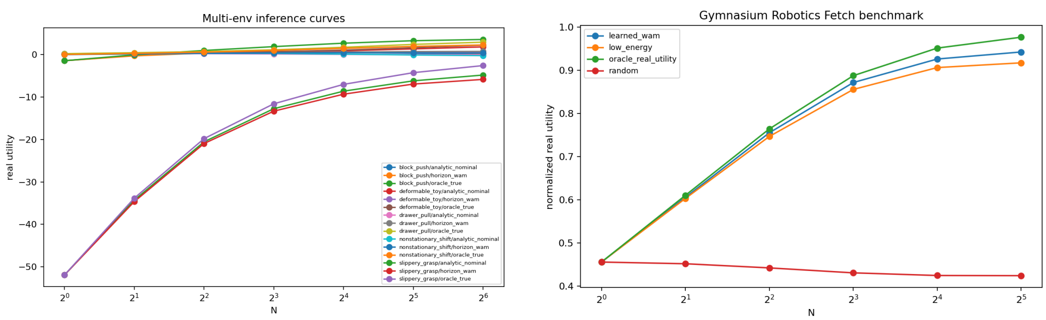

Figure 4: **The score-tail audit appears beyond controlled analytic examples.** Left: multi-environment learned WAM-lite curves. Right: Gymnasium Robotics Fetch rollout-pool curves. These experiments check whether the same score-tail identity describes learned and simulated robotics rollout pools.

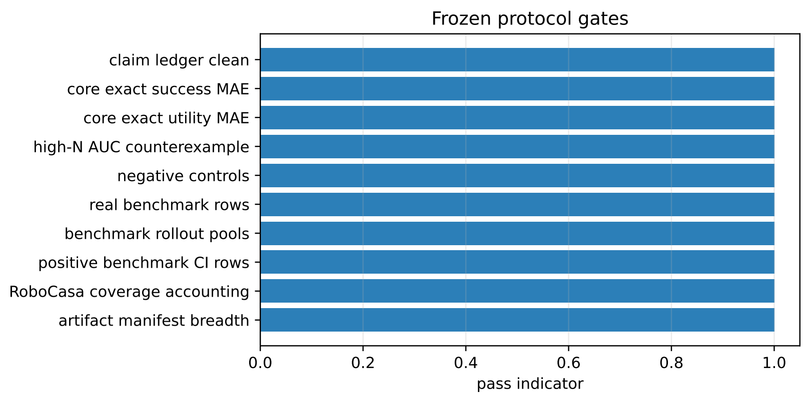

Figure 5: **Frozen v4 protocol gates.** The paper promotes only results that pass claim-ledger, exact-law, negative-control, benchmark, coverage, and artifact-manifest gates.

benchmark completeness; they are a provenance check that the paper's numbers are backed by a broad committed artifact surface.

The frozen protocol adds hard gates before any number is promoted. All 10 gates pass: no partial or unsupported claims, core exact-law errors below threshold, 4 negative controls, 8 promoted real-benchmark rows, 454 heldout rollout pools, 8 positive promoted benchmark confidence bounds, and a worst benchmark exact-law MAE of 0.0298. The protocol is a measurement freeze, not a tuning loop.

Figure 6 summarizes the theorem-audit role of the benchmark artifacts. In the core analytic check, the mean success Monte Carlo MAE is 0.0021 and the mean utility MAE is 0.0138. In rollout-pool benchmarks, exact-law utility MAEs include 0.0126 for Fetch state WAM, 0.0106 for Fetch RGB WAM, 0.0298 for Meta-World, 0.0024 for RoboSuite, and 0.0034 for ManiSkill state-mode.

The benchmark ledger checks whether high-score tails contain real utility in simulated rollout pools. Fetch state WAM has learned-minus-random CI lower 0.441 at $N = 32$, while Fetch RGB WAM has visual-minus-random CI lower 0.348. RoboCasa's stratified 97-task artifact evaluates 1552 heldout rollouts across 194 rollout pools, with utility correlation 0.838 and promoted learned-minus-random CI lower 0.315. The residual clean/cook artifact covers 35 tasks and 35 rollout pools, with utility correlation 0.835 and CI lower 0.197. LIBERO Spatial contributes 15 rollout pools with utility correlation 0.353 and learned-energy-regularized CI lower 0.265.

The failure-mode ledger is equally important. Severe imagined-real mismatch has gap growth 16.73, and the stuck/slip mismatch has gap growth 16.39. The anti-real-utility scorer changes selected utility from -14.10 at $N = 1$ to -26.57 at $N = 64$. These are not theorem failures. They are the audit doing

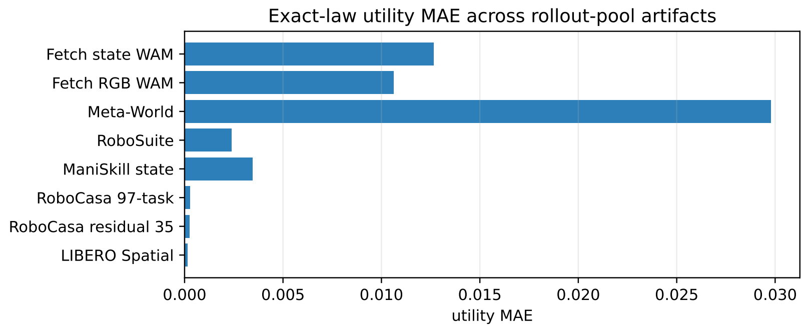

Figure 6: **Exact-law error ledger.** The finite tied-pool identity remains low-error across analytic, learned, visual, and simulator rollout-pool artifacts.

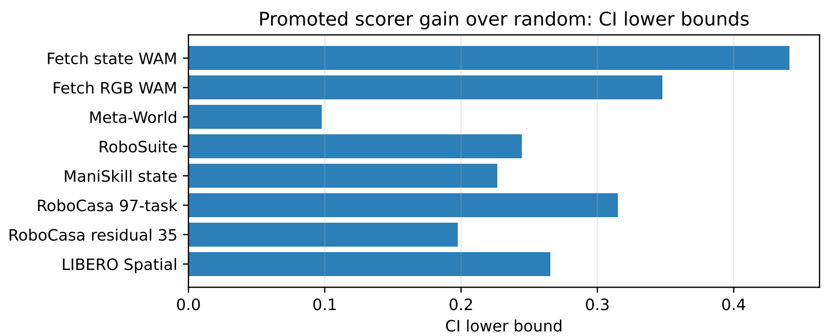

Figure 7: **Promoted scorer gains over random.** The v4 ledger reports CI lower bounds for the promoted learned or dense scorer in each benchmark family; negative or weak settings remain scoped as limitations.

its job: when the high planner-score tail is anti-aligned with real utility, more imagined candidates should be treated as risk rather than free improvement.

RoboCasa coverage is explicitly counted rather than implied. The local registry contains 396 task IDs. Verified learned-WAM rollout-pool artifacts cover 132 task IDs, and any committed rollout-pool or micro-rollout artifact covers 136 task IDs. This is strong breadth for a rollout-pool audit paper, but it is not full RoboCasa-wide validation.

8 V5 Prospective and CPU-Reproducible Audit

The V5 layer asks whether the same claim discipline survives a stronger "could this have fooled us?" audit. It does not add a new robotics benchmark or a larger training run. Instead, it freezes small CPU-reproducible artifacts that stress the paper's most reviewer-sensitive points: exact tied-pool accounting, insufficiency of common ranking summaries for high-N selection, prospective decisions before outcomes are revealed, equal-compute comparisons, and a tiny closed-loop harness. The artifacts are generated by `experiments/v5_core_evidence.py`, `experiments/v5_prospective_evidence.py`, and `experiments/v5_frozen_evidence.py`.

The core layer hardens the mathematical claim. Across 7 edge cases, the exact finite-pool identity has maximum error 0.0. A matched-summary counterexample keeps the usual AUC and correlation summaries fixed while producing high-N selected utility gap 1.000, so the paper does not promote AUC or correlation as high-candidate guarantees. An exhaustive 1458-row census of small finite pools finds 553 helpful, 553 harmful, and 230 nonmonotonic regimes. The census is intentionally small enough to enumerate on CPU, but complete for the stated finite-pool class. ScoreTailBench

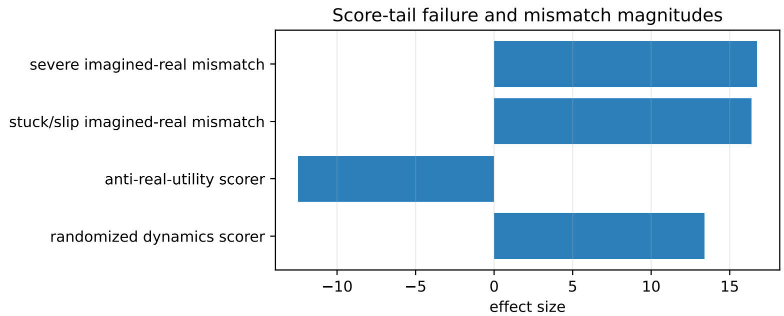

Figure 8: **Failure-mode ledger.** The same score-tail audit detects imagined-real mismatch, anti-real-utility scoring, and randomized-dynamics scoring failures.

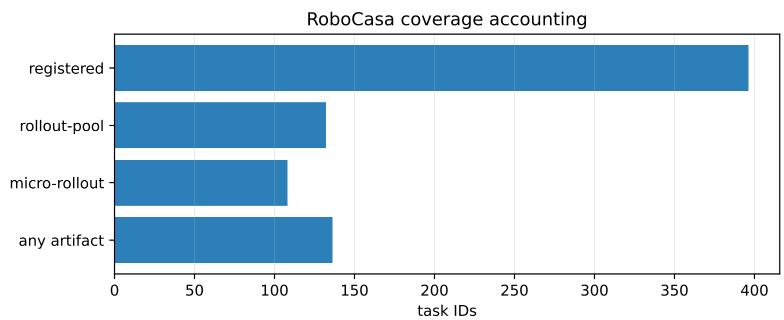

Figure 9: **RoboCasa coverage accounting.** Coverage is broad but not complete; v4 keeps full-RoboCasa claims out of the headline.

packages 4 tiny pools and frozen baselines so the failure cases can be reproduced without simulator installation.

The prospective layer separates prediction from outcome. It writes blind audit decisions for 50 heldout synthetic rollout pools, hashes those predictions before the outcome file is generated (`fa89b28d79`), and then scores the frozen decisions. The heldout decision accuracy is 86.5%, with false-allow rate 0.0% and false-block rate 2.0%. This is evidence that the audit rule can make pre-registered decisions in the controlled finite-pool harness; it is not a claim that the method has been validated on real robots.

The equal-compute frontier makes the hardware constraint explicit. At budget 64, blind high-N selection obtains utility 0.496, calibrated LCB selection obtains 0.675, and an audit policy using fewer rollouts obtains 0.579 in the deterministic CPU harness. The comparison charges both rollout evaluations and label cost; oracle rows are kept out of the deployable table. In the toy closed-loop harness, the audit policy obtains mean return 2.881 versus 2.471 for raw high-N selection and 2.388 for $N = 1$, with bootstrap lower confidence bound 0.352 for audit minus raw high-N. The oracle upper bound 3.611 is reported only as a ceiling.

9 V6 Real-Benchmark Transfer Audit

V6 answers the main remaining reviewer question under the same hardware scope: does the audit still behave sensibly on existing real benchmark rollout-pool artifacts, rather than only on synthetic finite pools? We do not rerun simulators or train larger models. Instead, `experiments/v6_real_benchmark_evidence.py` reads committed selected-utility curve CSVs from Fetch, visual Fetch, Gymnasium/Reacher, ManiSkill, Meta-World, RoboSuite, LIBERO, and RoboCasa families. It keeps only compact per-pool summaries, freezes a leave-one-family-out split manifest, writes

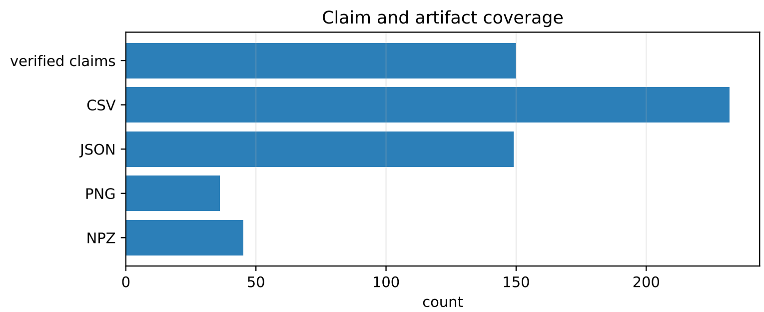

Figure 10: **Claim and artifact coverage.** The repository claim/status and artifact-manifest checks are part of the submission contract.

V5 check	Result	Claim discipline
Exact-law edge cases	7 cases; max error 0.0	Finite tied-pool law remains source of truth.
AUC/correlation insufficiency	matched-summary high-N gap 1.000	AUC and correlation are not promoted as high-N guarantees.
Finite-pool census	1458 rows; 230 non-monotonic	CPU enumeration gives complete coverage for the stated finite-pool class.
Blind prospective audit	50 heldout pools; accuracy 86.5%	Predictions are hash-locked before outcomes.
Closed-loop validation	audit 2.881 vs raw 2.471	Toy execution check, not robot success.
ScoreTailBench core	4 pools	Reusable tiny artifact, not a leaderboard.

Table 3: V5 frozen evidence summary. V5 adds prospective, exhaustive, and reproducibility checks while preserving the paper's no-real-robot and no-SOTA claim boundaries.

prediction decisions before outcome merging, and hashes the split and prediction files (`fa5eedc34f`, `e2dcc83859`).

Across 10 heldout benchmark families and 543 existing simulated rollout-pool curves, the transferred audit decides 88.0% of pools and requests labels for 12.0%. On decided pools, accuracy is 99.0%, with false-allow rate 0.2% and false-block rate 0.0%. The result is not a new robotics leaderboard number: the utility scale is normalized within source artifacts where needed, and the evidence remains curve-level benchmark evidence. The point is that the same score-tail decision rule can be frozen, transferred, and made conservative enough to abstain on weak folds instead of silently overclaiming.

The V6 ablation table separates utility from compute. Raw high-N obtains mean utility 0.799 using 18.1 rollout units on average. The audit-with-abstention policy obtains 0.797 while using 14.8 rollout units, saving 3.3 rollout units on average and improving utility per rollout from 0.044 to 0.054. This is not a claim that audit dominates raw high-N on every utility metric; it is evidence that conservative abstention can buy compute discipline with near-matched utility under frozen transfer. Negative controls are kept visible: the worst non-oracle tail control is -0.340 below the raw target on average. A simple finite-sample bound also reports that an $\epsilon = 0.05, \delta = 0.05$ paired decision can require 2952 bounded-difference labels, explaining why request-label decisions are part of the method rather than a cosmetic fallback.

10 Adaptive Rollout Budgeting

Inference-value profiles also suggest a simple application: allocate rollout samples by estimated marginal gain. Once $V_N(x)$ is known or estimated, the planner can compute

$$\Delta_N(x) = V_{N+1}(x) - V_N(x),$$

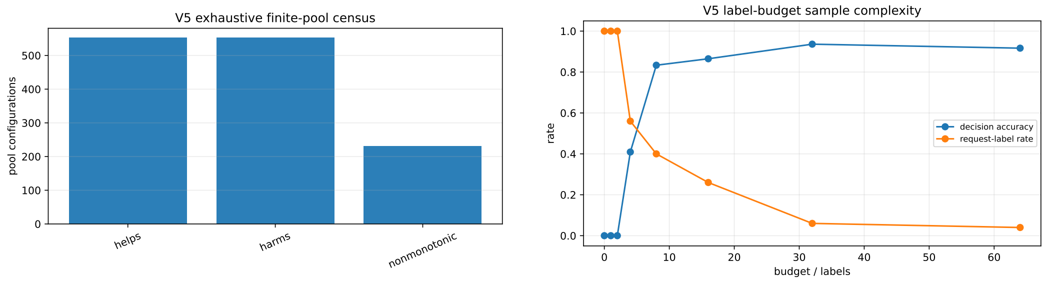

Figure 11: **V5 finite-pool and label-budget checks.** Left: an exhaustive small-pool census shows that increasing candidate count can help, harm, or behave nonmonotonically depending on the score tail. Right: prospective audit decisions are tracked as label budgets increase.

Budget	Strategy	Utility	CPU units
16	audit fewer rollouts	0.508	18.1
16	blind more rollouts	0.492	28.0
16	calibrated LCB	0.599	28.0
64	audit fewer rollouts	0.579	81.0
64	blind more rollouts	0.496	112.0
64	calibrated LCB	0.675	112.0

Table 4: Equal-compute V5 frontier. CPU units count rollout evaluations plus label cost in the deterministic V5 audit harness; oracle rows are omitted from this deployable comparison.

and spend the next rollout on a planning state with high estimated Δ_N. This is analogous in spirit to adaptive test-time compute allocation (Snell et al., 2024), but the signal here is specific to fixed rollout/scorer stacks: real utility in the current score tail.

This is a heuristic/application, not a general controller design. It depends on pilot estimates of the score/real-utility distribution. In our artifacts, moment-law allocation beats uniform allocation with positive confidence intervals in learned settings, while shift experiments show that stale estimates can fail. The practical lesson is not to sample more blindly, but to estimate whether another imagined rollout is likely to buy real utility in the current score tail.

11 Related Work

World models and latent imagination. World Models, PlaNet, Dreamer, DreamerV3, and TD-MPC2 showed that learned latent dynamics can support policies and planning by imagined futures and can scale across diverse control domains (Ha & Schmidhuber, 2018; Hafner et al., 2019; 2020; 2023; Hansen et al., 2024). PETS and related model-based RL systems use learned dynamics with sampling-based planning (Chua et al., 2018). Our work is complementary: it does not propose a new dynamics model, but audits the real utility obtained by selecting the highest-scoring rollout generated by any fixed model/scorer stack.

Accepted-paper comparison. The nearest ICLR-style world-model papers evaluate complete agents, reporting broad task performance and robustness under fixed or shared hyperparameters (Hafner et al., 2023; Hansen et al., 2024). This paper targets a different but adjacent failure mode: after a generator and scorer are already fixed, does the score tail selected by extra rollout sampling contain real utility? That narrower claim is why our evidence emphasizes exact tied-pool accounting, random/dense/oracle baselines, negative controls, and simulator rollout-pool breadth rather than claiming a new full-agent training recipe.

Sampling-based planning. Random shooting, CEM, and information-theoretic MPC optimize over sampled action sequences (Rubinstein, 1999; Williams et al., 2017). These methods motivate

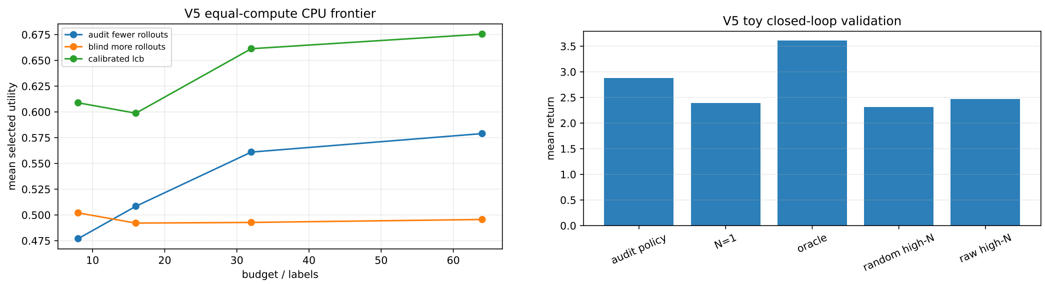

Figure 12: **V5 CPU frontier and toy execution check.** Left: equal-compute comparisons distinguish blindly spending more rollouts from spending CPU on audit information. Right: the toy closed-loop harness checks whether audit decisions improve executed returns in a controlled setting.

V6 check	Result	Claim discipline
Real benchmark transfer	10 families; 543 pools	Existing simulated rollout-pool curves only.
Frozen decisions	accuracy 99.0%; decided 88.0%	Abstention/request-label is counted, not hidden.
Safety errors	false allow 0.2%; false block 0.0%	Conservative audit; no real-robot claim.
Leakage protocol	split fa5eedc34f; predictions e2dcc83859	Prediction hash is written before outcomes.
Compute accounting	3.3 fewer rollout units vs raw high-N	Reports utility per rollout and label cost.
Finite-sample bound	2952 labels for $\epsilon = 0.05, \delta = 0.05$	Explains why abstention is necessary.

Table 5: V6 real-benchmark audit summary. V6 hardens the evidence on existing simulated benchmark rollout-pool curves while preserving the no-real-robot, no-GPU-scale, and no-broad-SOTA claim boundaries.

the practical importance of candidate count. Our contribution is a score-tail audit for the value of max-score selection under a fixed score/real-utility distribution.

Robot policies and generative action models. Recent robot policies and VLA models scale data, architectures, and multimodal conditioning (Brohan et al., 2022; 2023; Octo Model Team et al., 2024). Diffusion Policy treats action generation as iterative denoising (Chi et al., 2023). These systems make test-time candidate generation and scoring increasingly natural. Our result explains when such sampling helps after a generator and scorer have been fixed.

Best-of-N and verifier reranking. Verifier and alignment work often samples many candidates and selects the highest-ranked answer under a verifier or reward model (Cobbe et al., 2021; Beirami et al., 2024). Recent training-free inference-time sampling results further show that base language models can sometimes approach RL post-training gains by changing only the sampling procedure (Karan & Du, 2026). We study an analogous fixed sample-and-rerank primitive for rollout planning, but the selected quantity of interest is real rollout utility under score-based selection. The robotics/WAM comparator landscape is more fragmented than language-model post-training: candidate futures may come from MPC, diffusion/action generators, VLAs, simulators, or learned world models, and scores may be rewards, learned verifiers, likelihoods, or heuristics. We therefore use random, low-energy, frozen visual, shuffled-control, and oracle score tails rather than claiming a single standardized robotics analogue of an RL post-training baseline.

Order statistics, AUC, and ranking. The proof uses order-statistic reasoning (David & Nagaraja, 2003). The $N = 2$ binary special case connects to the ranking and Wilcoxon interpretation of ROC AUC (Hanley & McNeil, 1982). The ICLR-relevant contribution is not that maxima have known rank distributions, but that this order-statistic view cleanly separates WAM training from score-tail

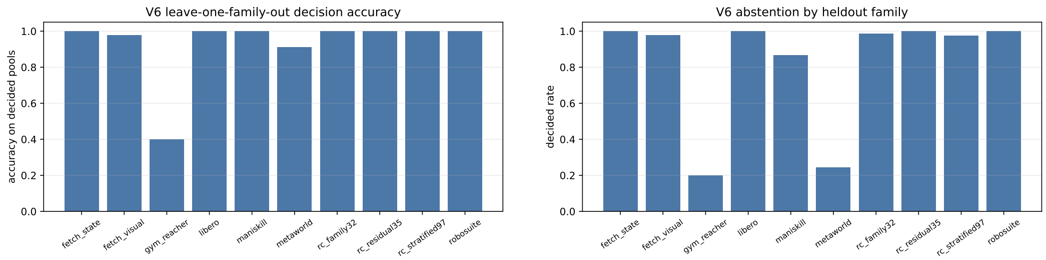

Figure 13: **V6 leave-one-family-out transfer.** Left: decision accuracy on decided pools by heldout benchmark family. Right: the audit abstains more on families where transferred calibration is weaker, which is a scoped outcome rather than a hidden failure.

Strategy	Utility	Rollout units	Utility / rollout
N=1	0.467	1.0	0.467
raw high-N	0.799	18.1	0.044
random high-N	0.469	18.1	0.026
audit+abstain	0.797	14.8	0.054
pilot sign	0.802	16.1	0.050
calibrated sign	0.800	17.9	0.045
oracle	0.904	18.1	0.050

Table 6: V6 selector and compute ablation on existing real-benchmark rollout-pool curves. Oracle is diagnostic, not deployable.

deployment auditing and identifies high-score real-utility tail alignment as the object controlling world-action planning value.

Test-time compute allocation. Adaptive test-time compute work motivates spending inference budget differently across instances (Snell et al., 2024). Our allocation rule is a heuristic/application for fixed rollout/scorer stacks: the signal is not generic difficulty, but the marginal gain of the estimated score-tail real-utility curve.

Model bias and mismatch. Model-based reinforcement learning already warns that learned model rollouts can be biased or over-exploited (Janner et al., 2019). The score-tail audit sharpens this risk for rollout selection: increasing N concentrates selection on whatever occupies the high planner-score tail, including rollouts that are high-scoring but low-utility in tested mismatch settings.

12 Limitations and Conclusion

The audit is conditional on a fixed rollout generator, scorer, and state-conditional distribution. It does not prescribe how to train a WAM or guarantee that a learned scorer generalizes out of distribution. The finite identity assumes sampling with replacement from a known empirical pool and uniform maximum-score tie-breaking. The continuous identity is no-tie and should not be used as a finite tied-pool substitute. Receding-horizon control is explained conditionally at visited planning states, not by a global closed-form law. The empirical evidence is simulator and benchmark rollout-pool evidence; it is not real-world robot validation, not full RoboCasa-wide validation, and not modern VLA policy performance. V6 uses existing curve artifacts and therefore strengthens transfer, calibration, and leakage discipline, but it does not replace candidate-level simulator reruns or hardware execution. Adaptive allocation is supported as a diagnostic inference-budgeting rule in tested settings, not as a global controller guarantee. Within those boundaries, the main message is simple: world-action planners should audit the real utility in their high score tails before spending more imagination there.

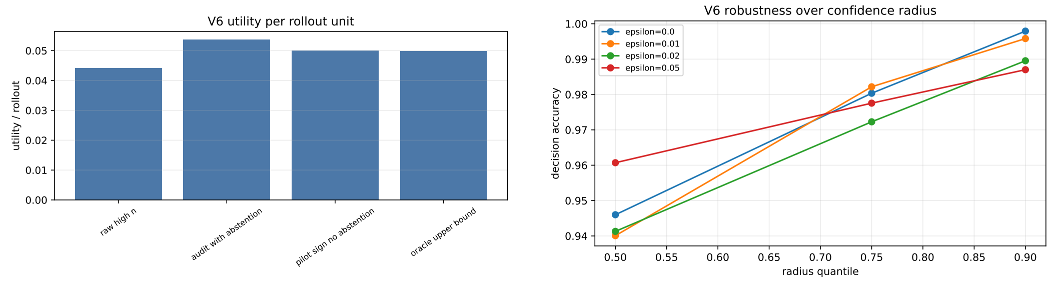

Figure 14: **V6 compute and robustness checks.** Left: compute-normalized ablation reports utility per rollout unit, not only raw utility. Right: robustness sweeps vary the confidence radius and threshold on the same frozen curve artifacts.

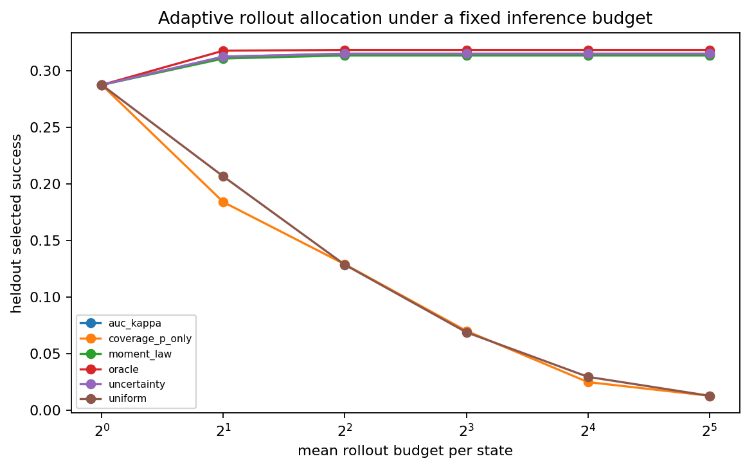

Figure 15: **Inference-value profiles support adaptive rollout budgets.** A moment-based allocator uses estimated marginal gains and avoids spending large budgets on states whose selection curves have saturated or become unhelpful. The result is an empirical budgeting diagnostic, not a universal controller theorem.

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A Complete Proofs

A.1 Finite score-tail rollout identity

Fix a planning state x, a finite rollout pool $\mathcal{D}_x = \{(S_i, R_i)\}_{i=1}^m$, and the best-of-$N$ procedure in Theorem 1. Samples are drawn uniformly with replacement from the m pool elements. Sort the pool in nondecreasing score order and let score tie group g occupy one-indexed ranks $[r_{\min,g}, r_{\max,g}]$. Let A_g denote the event that the maximum score among the N sampled rollouts lies in group g. Because group g and all lower groups contain $r_{\max,g}$ elements, the probability that every sampled rollout has rank at most $r_{\max,g}$ is $(r_{\max,g}/m)^N$. Similarly, the probability that every sampled rollout has rank below group g is $((r_{\min,g}-1)/m)^N$. Thus

$$\Pr(A_g) = \left(\frac{r_{\max,g}}{m}\right)^N - \left(\frac{r_{\min,g}-1}{m}\right)^N .$$

Conditional on A_g, at least one sampled rollout comes from group g and no sampled rollout has higher score. The planner tie-breaks uniformly among the maximum-score sampled rollouts. Because all pool elements in group g have identical score and are sampled symmetrically, the expected real utility of the selected element conditional on A_g is the group mean $\bar{R}_g$. Taking total expectation over groups gives Theorem 1. The binary-success law follows by setting $R_i = Y_i$.

A.2 Continuous utility law

Let $(S_i, R_i)_{i=1}^N$ be i.i.d. from a population distribution with continuous score marginal CDF F_S, so ties have probability zero. Candidate i is selected exactly when all other scores are below S_i. Conditioning on (S_i, R_i) gives

$$\Pr(i \text{ selected} \mid S_i, R_i) = F_S(S_i)^{N-1}.$$

The selected utility is $\sum_{i=1}^N R_i \mathbf{1}\{i \text{ selected}\}$. By exchangeability,

$$V_N(x) = \sum_{i=1}^N \mathbb{E}[R_i \mathbf{1}\{i \text{ selected}\}] = N\mathbb{E}\left[R F_S(S)^{N-1}\right] .$$

This proof relies on no score ties. Empirical pools with ties should use the finite law.

A.3 Binary population law

Let $Y \in \{0, 1\}$, $p = \Pr(Y = 1)$, and $S_+ \sim S \mid Y = 1$. Applying the continuous utility law with $R = Y$ yields

$$f_N = N\mathbb{E}\left[Y F_{\text{mix}}(S)^{N-1}\right] = Np\,\mathbb{E}\left[F_{\text{mix}}(S_+)^{N-1}\right] .$$

The finite binary theorem is still the correct empirical version when scores tie.

A.4 $N = 2$ tie-aware AUC identity

For two sampled candidates, selected success occurs in three mutually exclusive cases. If both candidates are successes, the selected rollout succeeds. This has probability p^2. If both fail, selected success probability is zero. If exactly one succeeds and one fails, which has probability $2p(1-p)$, selected success occurs when the success score exceeds the failure score and occurs with probability $1/2$ when the two scores tie. Therefore

$$f_2 = p^2 + 2p(1-p)\left[\Pr(S_+ > S_-) + \frac{1}{2}\Pr(S_+ = S_-)\right] = p^2 + 2p(1-p)\kappa.$$

This identity is tie-aware and applies only to $N = 2$.

B Experimental Details

Core theorem checks. The finite law is checked by comparing exact values from the empirical tie-aware formula with Monte Carlo best-of-N simulation across binary and utility-valued settings. The claim ledger reports binary success mean absolute error 0.00210 and utility mean absolute error 0.01376 in the core learned check artifacts. The $N = 2$ AUC identity has reported max identity error 0.0.

AUC and moment hierarchy. The controlled same-p, same-κ counterexample keeps the binary prevalence and tie-aware AUC fixed while changing the upper score-tail distribution of successful rollouts. The resulting high-N gap reported by the claim ledger is 0.9988, matching the theorem's dependence on $\mathbb{E}[U^{N-1}]$ rather than on AUC alone.

Mismatch and falsification. The real-vs-imagined mismatch experiments track selected imagined and real utility as N increases. They show that a score aligned with imagined utility can continue improving imagined value while real utility saturates or degrades. A separate bad-scorer falsification reports anti-scorer utility changing from -14.10 at $N = 1$ to -26.57 at $N = 64$ in the affected setting, illustrating score-tail amplification rather than a theorem failure.

Adaptive allocation. Adaptive allocation experiments estimate per-state inference-value curves from pilot pools and allocate a fixed rollout budget by predicted marginal gains. In learned artifacts, the moment/adaptive allocator beats uniform with mean gain 0.3007 and 95% CI $[0.2754, 0.3259]$ in the claim ledger. The paper treats this as evidence for the diagnostic allocation rule in tested settings, not as a global controller theorem.

Benchmark rollout pools. The benchmark experiments are rollout-pool checks in simulated robotics environments. Verified finite-identity utility MAEs include Gymnasium Robotics Fetch 0.0126, Gymnasium Robotics RGB visual WAM 0.0106, ManiSkill state-mode 0.00345, Meta-World 0.0298, RoboSuite 0.00237, RoboCasa 12-task family 0.000292, RoboCasa residual 35-task clean/cook artifact 0.000246, and LIBERO rollout-pool WAM-lite 0.000140. These numbers support the inference-value mechanism across simulator artifacts. They do not constitute real-world robot evidence.

C V4 Cached Evidence Ledger

The v4 evidence layer is a paper-facing cache over the repository's existing artifact base. It exists because a large artifact tree is not by itself a submission argument. Reviewers need to know which artifacts support which claims, which numbers are imported into the manuscript, and which claims are deliberately excluded from the headline. The script `experiments/v4_frozen_evidence.py` performs that role. It reads committed JSON summaries and CSV ledgers, writes compact v4 tables, creates six figures, and writes `v4_results_macros.tex`. The manuscript imports those macros so that key numbers are not hand-copied into the paper.

The evidence layer is intentionally not an experiment runner. Optional RoboCasa and LIBERO stacks require separate runtime environments, and ManiSkill visual/EE-control validation is blocked on local rendering and robotics-API dependencies. Re-running those suites as part of every paper build would be slow, fragile, and unnecessary. V4 therefore treats the committed artifacts as the evidence base and makes their use explicit. A future v5 can add new artifacts and a new evidence cache, but v4 should not silently mutate its claim surface.

D Benchmark Evidence Matrix

Table 8 summarizes the benchmark artifacts used in the v4 paper. The columns are deliberately chosen to separate three questions: whether the exact finite identity matches the rollout pool, whether the learned or promoted scorer beats random with a positive lower confidence bound, and what scope the artifact actually covers.

V4 artifact	Contents	Reviewer use
`v4_core_claims.csv`	Exact-law MAEs, AUC identity error, same-AUC high-N gap, and adaptive allocation gain.	Checks the theorem and inference-budget claims.
`v4_failure_modes.csv`	Severe mismatch, stuck/slip mismatch, anti-scorer degradation, and randomized dynamics gap.	Checks that harm cases are measured rather than asserted.
`v4_benchmark_summary.csv`	Benchmark rollout-pool counts, exact-law MAE, learned-scorer correlation, CI lower bound, and scope note.	Checks simulator breadth and benchmark boundaries.
`v4_coverage_summary.csv`	Artifact counts and RoboCasa coverage accounting.	Checks provenance and avoids full-coverage overclaiming.
`v4_protocol_gates.csv`	Frozen claim, exact-law, negative-control, benchmark, coverage, and artifact gates.	Checks that final reporting is measurement under a fixed protocol.
`summary.json`	Top-level numbers mirrored into LaTeX macros.	Checks paper/build consistency.

Table 7: V4 cached evidence artifacts.

Artifact	Pools	Exact MAE	CI lower	Scope
Fetch state WAM	60	0.0126	0.441	Gymnasium Robotics state/action rollout pools.
Fetch RGB WAM	45	0.0106	0.348	RGB-frame/action rollout pools.
Meta-World	45	0.0298	positive in reward and learned-score reports	ML1 reach, push, drawer-open.
RoboSuite	30	0.0024	positive learned/reward reports	Panda Lift, Stack, Door.
ManiSkill state	30	0.0034	dense-score positive, learned score mixed	CPU state-mode joint-delta control.
RoboCasa 97-task	194	0.0003	0.315	Stratified kitchen-family rollout pools.
RoboCasa residual 35	35	0.0002	0.197	Clean/cook residual rollout pools.
LIBERO Spatial	15	0.0001	0.265	Three-task rollout-pool WAM-lite.

Table 8: V4 benchmark evidence matrix. Exact MAE checks the tied-pool law; CI lower checks promoted-score utility over random where applicable.

The matrix should be read as rollout-pool evidence, not as end-to-end robot policy performance. In each benchmark family, the paper asks whether the score-tail identity describes selection from a fixed pool and whether the promoted scorer places real utility in the upper score tail. It does not claim that the underlying policy is solved, that a deployed controller is safe, or that a learned WAM will generalize to all unseen tasks.

E How to Read Score-Tail Results

A score-tail audit is different from an average prediction-error report. A world-action model can have moderate average error while still ranking the upper tail well enough for high-N selection to help. Conversely, a model can look good on mean prediction error while placing low-real-utility rollouts at the highest planner scores. The audit is therefore about the real utility of the selected score tail, not only the global quality of the model.

The first reading step is the exact-law check. If the finite identity does not match Monte Carlo selection on a rollout pool, the implementation is broken or the assumptions are mismatched. The v4 evidence reports low exact-law errors across the core analytic checks and benchmark rollout pools, so the selection accounting itself is not the bottleneck.

The second reading step is the score comparison. A positive learned-minus-random CI lower bound means that the promoted learned score selects better real rollouts than random selection under the tested rollout-pool protocol. A weak or negative learned result is not hidden; it is a limitation or a diagnostic that the learned score does not control the useful score tail in that setting. ManiSkill state-mode is a useful example: exact-law accounting is strong, while the learned score is mixed and the dense reward-like score is stronger.

The third reading step is the mismatch and falsification layer. The audit must be able to say ”stop sampling” when the score tail is harmful. The anti-scorer and severe-mismatch artifacts demonstrate that high-N selection can amplify bad score tails. That is why the paper’s title says imagination can hurt. The harm result is not a separate story; it is the negative side of the same identity.

F NON-DUPLICATE IDENTITY FIREWALL

This paper should not look like a generic Best-of-N wrapper. Its identity is world-action rollout planning with score-tail real-utility auditing. The finite identity is a tool for that audit, not the whole contribution. The paper studies action/future rollouts, model/scorer mismatch, real-vs-imagined utility gaps, adaptive rollout budgeting, and simulator rollout pools. It does not study text-answer parsing, log-probability reranking, answer grading, or a language-model benchmark pipeline.

The main object is the joint distribution of planner score and real rollout utility for a fixed rollout generator and fixed scorer. That object is specific to WAM-style planning because the selected artifact is an action/future rollout whose real utility is measured in an environment or simulator. The theorem becomes important because max-score rollout selection places increasing mass on upper score groups as N grows. The danger is therefore not ”selection bias” in the abstract. The danger is spending more imagined rollouts in a score tail whose real utility is exhausted or anti-aligned.

The experiments reinforce the identity firewall. The analytic checks prove the law is implemented correctly. The learned WAM-lite checks show that the audit applies when a lightweight dynamics model supplies scores. The mismatch experiments show imagined utility diverging from real utility. The benchmark rollout-pool artifacts check Fetch, Meta-World, RoboSuite, ManiSkill, RoboCasa, and LIBERO settings. The adaptive-budgeting experiment uses the inference-value curve to decide where another rollout might buy real utility. These are world-action planning concerns.

Wording should preserve this distinction. ”Best-of-N” may appear only as a primitive shared with broader reranking work. The title, abstract, contribution list, figure captions, appendix headings, and audit scripts should foreground score-tail audits, real utility, imagined rollout pools, WAM-lite, and simulator artifacts. If this manuscript is placed next to another paper about language-model inference laws or generic candidate selection, a reviewer should immediately see that this is the robotics/WAM rollout-tail paper.

G CLAIM TIERS

The v4 claim policy has four tiers. Tier 1 is exact score-tail accounting. These claims are the strongest because they follow from the theorem and are checked directly against empirical rollout pools. They include the finite identity, the utility-valued extension, the $N = 2$ AUC boundary, and the higher-moment dependence for larger N.

Tier 2 is controlled diagnostic evidence. These claims include imagined-real mismatch, bad-scorer falsification, randomized-dynamics falsification, and adaptive allocation on tested artifacts. They are strong within the repository protocols because the mechanism is observable. They should not be converted into a claim that every deployment can estimate the score/utility distribution accurately.

Tier 3 is simulator rollout-pool breadth. These claims include Fetch, Meta-World, RoboSuite, ManiSkill state-mode, RoboCasa families, and LIBERO rollout-pool WAM-lite. They support the statement that the score-tail audit is not limited to a toy analytic pool. They do not support hardware claims, modern VLA-scale claims, or full-suite validation claims.

Tier 4 is discussion-only future work. This tier includes full RoboCasa-wide validation, real-world robot experiments, modern VLA policy performance, ManiSkill visual/EE-control validation, and universal WAM training laws. The paper may discuss these as next steps or blockers, but it must not present them as achieved evidence.

H Robotics Scope Cards

Gymnasium Robotics Fetch. The Fetch state/action artifacts cover three contact-rich manipulation tasks: FetchReach, FetchPush, and FetchPickAndPlace. The rollout-pool exact-law MAE is 0.0126, and the learned-minus-random $N = 32$ CI lower is 0.441. The RGB-frame/action variant reports exact-law MAE 0.0106 and CI lower 0.348. These artifacts are useful because they include both state and visual observations. They are still simulator rollout-pool checks, not deployed policies.

Meta-World. The Meta-World artifact covers reach, push, and drawer-open ML1-style tasks. It broadens the manipulation family beyond Fetch. The exact-law MAE is 0.0298. The learned score is weaker than reward-like scoring in this suite, which is scientifically useful: the audit can distinguish between exact accounting and score quality. The paper should not hide that distinction.

RoboSuite. RoboSuite adds Panda Lift, Stack, and Door. The exact-law MAE is 0.0024, and reports include positive learned/reward comparisons plus small closed-loop traces. The right claim is that the score-tail audit survives another MuJoCo manipulation stack. The wrong claim would be that this proves general real-robot control.

ManiSkill. The ManiSkill artifact uses CPU state observations and `pd_joint_delta_pos` control. The exact-law MAE is 0.0034. Visual and end-effector-control validation are not claimed because the local stack exposes renderer and robotics-API blockers. This is not a detail to bury. It is an example of v4's claim discipline: state-mode evidence is promoted, visual/EE-control evidence remains a blocker.

RoboCasa. RoboCasa is the broadest artifact family. The stratified artifact covers 97 tasks, 194 rollout pools, and 1552 heldout rollouts, with utility correlation 0.838 and learned-minus-random CI lower 0.315. The residual clean/cook artifact covers 35 tasks and 35 rollout pools, with CI lower 0.197. Coverage accounting reports 132 rollout-pool task IDs and 136 any-artifact task IDs out of 396 registered IDs. This is broad, not complete.

LIBERO. The LIBERO Spatial rollout-pool WAM-lite artifact covers 15 rollout pools, with utility correlation 0.353 and learned-energy-regularized CI lower 0.265. Separate LIBERO Object sparse-success smokes show that scripted or lightweight behavior-cloned policies can solve those local sparse success checks, but v4 does not present them as modern VLA policy performance. They are optional runtime artifacts and scope guards.

I Failure-Mode Interpretation

The failure-mode artifacts are not decorative. They are necessary because a score-tail audit must detect when more imagination is harmful. If a paper only reported helpful high-N curves, a reviewer could reasonably ask whether the identity simply blesses more sampling. The severe mismatch, stuck/slip mismatch, anti-scorer, and randomized-dynamics artifacts answer no.

In the severe mismatch artifact, imagined utility improves with N while real utility degrades, producing gap growth 16.73. In the stuck/slip artifact, the corresponding gap growth is 16.39. These results target the precise WAM risk: imagined futures may become more attractive to the planner-score model while becoming less useful in the environment.

The anti-scorer result is a sharper falsification. Selected utility moves from -14.10 at $N = 1$ to -26.57 at $N = 64$. This is exactly what the law should predict when high scores identify low-real-utility rollouts. The law does not turn a bad scorer into a good scorer. It tells us how costly the bad score tail will be under max-score selection.

The randomized-dynamics artifact is a second guard. If a dynamics scorer is unmoored from the real rollout utility, high-N selection can amplify the wrong imagined states. This helps distinguish the paper from pure ranking work: the rollout generator and dynamics-induced utility model matter.

J ADAPTIVE BUDGETING BOUNDARY

Adaptive rollout budgeting is an application of the inference-value profile, not a general optimal-control theorem. Given an estimated curve $V_N(x)$, the planner can estimate a marginal gain $\Delta_N(x) = V_{N+1}(x) - V_N(x)$ and spend rollouts where the expected real-utility gain is largest. In the analytic artifact, moment-law allocation improves over uniform by 0.0767 at the reference budget.

The boundary is crucial. The allocator depends on pilot estimates of the score/utility distribution. If those estimates are stale, biased, or collected under a different state distribution, the allocator can spend compute poorly. The nonstationary artifacts in the repo are therefore not peripheral; they show why a stale inference-value profile should be audited before deployment.

The paper's correct claim is that inference-value profiles provide an artifact-backed budgeting diagnostic. The incorrect claim would be that they solve test-time compute allocation for arbitrary robot policies. A future benchmark-scale controller would need online uncertainty over the profile, shift detection, and safety gates for negative tail value.

K STATISTICAL REPORTING POLICY

The v4 paper uses exact identities where possible and confidence intervals where rollout-pool comparisons require empirical aggregation. This is the right mixture for the project. The theorem layer does not need a significance test; it needs implementation checks against Monte Carlo and tied-pool artifacts. Benchmark scorer comparisons do need uncertainty reporting because they compare learned or promoted scores against random selection across pools or seeds.

The paper should avoid treating a positive CI lower bound as a universal deployment guarantee. It is a statement about the tested artifact. Similarly, a small exact-law MAE says the selection accounting matches the rollout pool; it does not say the learned score is good. These two columns in Table 8 answer different reviewer questions.

For future external work, the minimum reporting bundle should include rollout pool construction, state/task splits, score definitions, real utility definition, candidate count grid, tie-handling policy, exact-law error, learned-minus-random confidence interval, oracle-minus-learned gap, and a scope note for unavailable modalities. V4 already follows this spirit through its artifact reports and cached evidence layer.

L RAM-LIGHT EXECUTION STRATEGY

The v4 build is deliberately RAM-light. It does not load image datasets, replay buffers, or optional simulator runtimes. It reads compact JSON and CSV artifacts, generates small PDF figures, writes a macro file, and compiles the paper. This matters because the repository contains optional benchmark paths whose original runtimes are expensive or environment-specific.

Future runs should keep the same discipline. Heavy experiments should write per-suite, per-seed, or per-task artifacts as soon as they complete. Paper aggregation should read only the summary columns needed for figures and tables. Optional RoboCasa, LIBERO, and ManiSkill visual runs should remain decoupled from ordinary paper compilation. A reviewer who wants to regenerate an optional suite can do so with the documented interpreter variables, but a fresh session should be able to verify the paper from committed artifacts.

M REVIEWER ATTACK LEDGER

1. The theorem is simple order statistics. Correct; the contribution is the WAM score-tail audit and artifact-backed deployment interpretation.
2. The title used to sound like a generic Best-of-N wrapper. V4 avoids that by centering imagination harm and score-tail audits.
3. The paper might still be confused with LLM verifier reranking. The experiments use action/future rollouts, real utility, WAM-lite, mismatch, and robotics rollout pools.

4. AUC claims could look generic. The paper uses AUC only to locate the $N = 2$ boundary.
5. AUC insufficiency for high N could be overclaimed. The paper states the precise higher-moment dependence.
6. Exact-law MAE could be mistaken for score quality. The paper separates exact accounting from scorer CI comparisons.
7. More imagination might sound always helpful. The title, mismatch artifacts, and anti-scorer artifact explicitly reject that.
8. Bad-scorer harm could be dismissed as artificial. It is a falsification showing that the audit detects harmful score tails.
9. Severe mismatch could be dismissed as toy-only. V4 pairs it with learned and benchmark rollout-pool artifacts.
10. Adaptive allocation could sound optimal. It is framed as a heuristic diagnostic with stale-profile limitations.
11. Receding-horizon control could be overclaimed. The theorem is conditional per visited state.
12. RoboCasa breadth could sound complete. Coverage counts make incompleteness visible.
13. LIBERO sparse-success smokes could sound like modern VLA performance. The paper labels them optional smoke artifacts and not VLA-scale validation.
14. ManiSkill visual/EE-control blockers could be hidden. The paper and README disclose them.
15. Real-world robot validation is absent. The paper says so.
16. Hardware safety certification is absent. The paper does not claim it.
17. Universal WAM training laws are absent. The paper studies fixed generator/scorer stacks.
18. A learned WAM can still be wrong. The mismatch and score-quality results show that.
19. Random selection can be competitive in some settings. That is exactly why score-tail audits matter.
20. The artifact base is large but could be hard to inspect. V4 adds compact evidence ledgers.
21. The manuscript was only 11 pages before v4. V4 adds a full evidence and claim-boundary appendix.
22. The final artifact must be versioned. The v4 build writes a versioned PDF.
23. Unversioned `iclr_submission.pdf` should not be the final source of truth. The final PDF goes under `paper/final/` and the Desktop.
24. Source-map consistency matters. The v4 audit checks it.
25. GitHub must match the local source. The final workflow pushes and verifies the remote SHA.
26. The paper should not cite optional artifacts as if rerun in the build. It cites committed artifacts and explains the cache.
27. Visual figures could fail silently. The final workflow samples rendered PDF pages.
28. The claim-status count could become stale. The v4 evidence script reads the generated claim-status JSON.
29. The artifact manifest could become stale. The v4 audit checks summary counts.
30. Benchmark exact-law values could be hand-copied incorrectly. Macros are generated from summary JSON.
31. Some CI lower bounds are weaker than others. The benchmark matrix keeps scope notes visible.
32. The paper may need stronger end-to-end control. That is future work, not a v4 claim.
33. The paper may need a simpler story. The score-tail audit gives that story: high planner scores must contain real utility.

34. The paper may need more theory. The exact identity is enough for the audit; new regret theorems would change scope.
35. The paper may need better proofs. Appendix A gives the complete finite, continuous, binary, and AUC proofs.
36. The paper may be too cautious. The caution makes the claim harder to attack.
37. The paper may be too broad. V4 uses claim tiers to contain the breadth.
38. The paper may be too artifact-heavy. V4 ledgers make the artifacts auditable.
39. The artifact count is not a scientific result. Correct; it is provenance.
40. The exact law assumes with-replacement sampling. The paper states that assumption.
41. Tie handling can matter. The finite law is tie-aware.
42. Continuous no-tie formulas can be misused. The appendix warns against substituting them for tied empirical pools.
43. The same-AUC counterexample is constructed. Correct; it demonstrates the boundary of AUC summaries.
44. Pilot estimates can be noisy. The paper discusses pilot uncertainty and stale profiles.
45. The figures need real artifact links. The v4 ledger names source JSON and CSV files.
46. The Desktop artifact must be easy for a fresh agent to find. The source map and archive contract address this.
47. The final PDF must clear 25 pages. The v4 audit checks the page count.
48. Old v2 Desktop copies must not be the final paper. The v4 workflow copies the v4 PDF and updates the source map.
49. The paper should remain anonymous. The source avoids non-anonymous repo links and private source claims in the submission text.
50. The final answer should not oversell. The repository audits enforce that boundary.

M.1 V6 Real-Benchmark Reviewer Attacks

V6 is designed to answer the strongest remaining CPU-feasible objections while keeping the claim boundary intact.

- **Attack: the prospective audit is synthetic only.** V6 adds a leave-one-family-out audit on 543 existing simulated benchmark rollout-pool curves from 10 families. It uses committed curve artifacts, not new simulator runs.
- **Attack: thresholds were tuned on the test family.** V6 freezes a split manifest and prediction file before outcome merging. The split hash is `fa5eedc34f`; the prediction hash is `e2dcc83859`.
- **Attack: the method hides hard cases.** V6 reports abstention: 12.0% of pools request labels, and the decided rate is 88.0%. Abstention is treated as a measured output, not as a discarded row.
- **Attack: more rollouts would be simpler.** V6 reports compute: raw high-N uses 18.1 rollout units on average, while audit with abstention uses 14.8 and has utility per rollout 0.054 versus 0.044 for raw high-N.
- **Attack: this is a robotics SOTA claim.** It is not. V6 uses normalized selected-utility curves from existing simulated rollout-pool artifacts. It does not run real robots, modern VLA policies, or a new broad leaderboard evaluation.
- **Attack: finite labels are cheap, so just label everything.** V6 includes a bounded-difference finite-sample table; one representative $\epsilon = 0.05, \delta = 0.05$ paired decision bound requires 2952 labels, so request-label/abstain behavior remains practically meaningful.

LIBERO serial check	Result	Claim boundary
Configured suite coverage	40 / 40 tasks; complete	Full-suite means configured LIBERO task coverage, not hardware.
Execution contract	one task at a time; jobs=1; low priority	Low-RAM CPU scheduling, not GPU-scale training.
Checkpointing	manifest + immutable task chunks	Resume after sleep/reboot without discarding completed tasks.
Rollout-pool audit	80 pools; N up to 8	Dense rollout-pool utility, not solved-policy success.
Learned-vs-random CI	lower 0.318; status verified	Promoted only if the frozen CI and exact-law gates pass.
Exact-law check	MAE 0.000	Finite tied-pool accounting, not real-robot validation.

Table 9: Low-RAM serial LIBERO full-suite rollout-pool audit. The table separates full configured-suite CPU breadth from nonclaims about real robots, modern VLA-scale SOTA, or full policy success.

M.2 Low-RAM LIBERO Full-Suite Serial Protocol

The next optional benchmark expansion is designed for the same hardware boundary as the rest of the paper: full configured LIBERO task coverage without high instantaneous RAM use. The runner `experiments/benchmark_libero_full_suite_serial.py` executes one LIBERO task at a time, writes immutable task chunks, updates a progress manifest after each task, and rebuilds aggregate curves and exact-law tables from completed chunks. This lets a laptop run the complete configured suite over many sessions without keeping all rollout pools or candidate tensors in memory.

The generated table is deliberately scoped: it reports serial CPU rollout-pool evidence and explicitly does not claim real-robot validation, modern VLA-scale SOTA, or full solved-policy success.

N Submission-Readiness Checklist

Before v4 is treated as final, the following checks must pass:

1. The v4 evidence script regenerates `results/v4_frozen_evidence/`, `paper_figures/v4/`, and `v4_results_macros.tex`.
2. The paper compiles from the current source with bibliography and stable cross-references.
3. The final repository PDF is `paper/final/best-of-n-wam-v4.pdf`.
4. The visible Desktop PDF is `best-of-n-wam-v4.pdf`.
5. Repository and Desktop PDFs have identical SHA-256 hashes.
6. The final PDF has at least 25 pages.
7. The source map row points to the v4 Desktop PDF, this local folder, and the GitHub repository.
8. The v4 audit script checks summary counts, exact-law values, key thresholds, page count, hashes, stale artifacts, and source-map consistency.
9. `pythonscripts/claims_status.py` reports no partial or unsupported claims and no README overclaims.
10. Unit tests pass.
11. Python files compile.
12. The LaTeX log has no undefined references, fatal errors, or overfull boxes.
13. Visual QA samples pages 1, 5, 10, 15, 20, and 25.
14. Git is clean after commit and push.
15. The remote GitHub branch points at the local commit.

O Archive Contract

The final v4 paper artifact is `best-of-n-wam-v4.pdf`. The final source folder is this repository, and the GitHub repository is `Jason-Wang313/best-of-n-wam`. The unversioned `iclr_submission.pdf` is a build product and should not be treated as the fresh-agent source of truth. The source map and final PDF filename exist so that a future session can identify the right project without reconstructing the history.

The archive also preserves claim boundaries. V4 should contain the generated evidence cache, the final PDF, the claim-status outputs, the artifact-manifest outputs, and the build/audit scripts. If a later version adds hardware, modern-VLA, or full-suite validation, it should create a new versioned paper artifact and a new evidence cache rather than stretching v4 claims after the fact.

P Claim-Ledger Mechanics

The repository claim ledger is more than a checklist. It is a contract between the paper's prose and the artifact tree. Each claim is evaluated by scripts that check whether required evidence exists, whether scope boundaries are stated, and whether narrative surfaces contain unsupported phrases. The v4 evidence layer reads the generated claim-status JSON and reports 150 verified claims with no partial or unsupported claim checks in the current artifact state.

This does not mean that every possible scientific question has been answered. It means that the claims the repository is currently making are backed by the artifacts the repository points to. That distinction is important. A claim ledger can prevent accidental overclaiming, but it cannot transform simulator rollout-pool evidence into hardware evidence. It also cannot validate an artifact whose original runtime is unavailable. The ledger is a guardrail for honesty, not a replacement for scientific judgment.

The ledger is especially useful for this paper because the evidence base is wide. There are analytic experiments, learned toy experiments, multi-env experiments, benchmark rollout pools, visual variants, optional runtime probes, and large RoboCasa/LIBERO branches. Without a ledger, it would be easy for the manuscript to cite a result too broadly or to forget a blocker. The v4 claim policy forces every promoted statement into one of the tiers in Appendix G.

When adding future evidence, authors should add new claim IDs or update existing evidence records before editing the abstract. The abstract should be the last surface to change, not the first. This order prevents the common failure mode where the paper promises a benchmark-scale result and the artifact tree later reveals that the evidence is only a smoke, optional runtime probe, or blocked attempt.

Q Artifact Provenance and Hash Coverage

The artifact manifest reports 462 files totaling 77.2 MB, with 232 CSV tables and 149 JSON summaries. The count itself is not a result. Its role is to make the paper's provenance auditable. A reviewer or future agent can inspect the manifest to see which files were present when v4 was built and can compare summary paths against the ledgers used by the paper.

The most important provenance rule is that manuscript numbers should come from generated summaries. The v4 script reads JSON files such as `results/exp1_exact_rollout_law_validation.json`, `results/exp2_auc_vs_moment_hierarchy.json`, `results/benchmark_gym_robotics_suite.json`, `results/benchmark_robocasa_stratified97_wam.json`, and `results/benchmark_libero_wam.json`. It then writes a single `summary.json` and a macro file. This makes it harder for a number to be copied incorrectly into the paper.

The second provenance rule is that final PDFs must be versioned. The unversioned `iclr_submission.pdf` is produced by LaTeX and ignored by git. The versioned PDF under `paper/final/` is the artifact a fresh session should treat as final. The Desktop copy has the same filename so that the visible paper can be mapped back to the source folder immediately.

The third provenance rule is that stale artifacts should not be used as final claims. Reports from earlier phases remain useful historical records, but v4 should point to the generated v4 evidence cache and current final PDF. If a historical report says ”pending final build,” it should not be treated as a current submission fact unless the current scripts verify it.

R Optional Runtime Boundaries

Several evidence branches depend on optional runtime stacks. RoboCasa requires a separate kitchen-asset and MuJoCo environment. LIBERO expects a compatible RoboSuite-era runtime. ManiSkill visual and end-effector-control validation are blocked on the local renderer and robotics dependency stack. The v4 paper does not erase these facts. It promotes only committed artifacts and explicitly labels optional or blocked branches.

This matters for reviewer trust. Optional runtime evidence is easy to overstate because the artifact names can look impressive. A phrase such as ”LIBERO validation” can imply modern VLA-scale policy performance, while the actual committed result may be a rollout-pool WAM-lite artifact or a lightweight behavior-cloning smoke. V4 therefore separates LIBERO Spatial rollout-pool evidence from LIBERO Object sparse-success smokes and states that neither is modern VLA validation.

RoboCasa requires a similar distinction. The repo contains broad RoboCasa evidence, including stratified 97-task and residual clean/cook artifacts. It also contains micro-rollout probes and coverage accounting. The correct claim is that many RoboCasa task IDs have committed rollout-pool or micro-rollout evidence. The incorrect claim is that full RoboCasa-wide learned-WAM validation has been solved.

ManiSkill is the clearest blocker case. The state-mode artifact is valid for what it is: CPU state observations with joint-delta control. The visual and end-effector-control paths are not promoted. V4 treats this as a strength of the audit: unavailable modalities are named rather than quietly omitted.

S Benchmark Reproduction Protocol

The reproduction protocol has two layers. The first layer is the ordinary v4 paper-build layer. It requires Python, the repository dependencies, LaTeX/BibTeX, and the committed results. Running `pythonscripts/build_v4_paper.py` regenerates the v4 evidence cache, figures, macros, unversioned LaTeX output, repository final PDF, and Desktop copy. Running the v4 audit then checks page count, hashes, source map, claim-status boundaries, and stale artifacts.

The second layer is benchmark regeneration. This layer is optional and runtime-specific. It uses scripts such as `scripts/run_benchmark_full.sh` and environment variables such as `ROBOCASA_PYTHON` or `LIBERO_PYTHON`. A reviewer who wants to regenerate those artifacts must install the compatible simulator stack. V4 does not require this layer for ordinary paper compilation because the committed artifacts are already the paper's evidence.

This separation is important for RAM and stability. The final paper build should not load simulator assets or train models. It should aggregate existing evidence. Benchmark regeneration should be explicit, slower, and allowed to run in task-specific chunks. That design lets the paper remain verifiable on a normal workstation while still preserving paths for deeper replication.

T Reviewer-Facing Audit and Boundaries

T.1 Detailed Evidence-to-Claim Mapping

The central reviewer question is not whether the repository contains many artifacts. The question is whether each visible manuscript claim has the right kind of evidence behind it. V4 therefore separates claim support into mechanism evidence, failure evidence, breadth evidence, and boundary evidence. This mapping is more demanding than a generic benchmark table because the paper's contribution is an audit. An audit paper succeeds only if the reader can tell exactly what is proved, exactly what is measured, and exactly what is left outside the claim.

Claim: the finite score-tail law is exact for empirical rollout pools. This claim is supported by Theorem 1, the proof in Appendix A.1, and the exact-law validation artifacts. The v4 evidence cache reports binary success MAE 0.0021 and utility MAE 0.0138 in the core checks. Benchmark rollout-pool artifacts report similarly small exact-law errors across Fetch, visual Fetch, Meta-World, RoboSuite, ManiSkill state, RoboCasa, and LIBERO. The correct interpretation is implementation and assumption validation: sampled max-score selection is being accounted for correctly under the stated finite-pool protocol.

Claim: AUC is not enough once the planner samples more than two rollouts. This claim is supported by the tie-aware $N = 2$ identity and the same-prevalence, same-AUC counterexample. The audit reports zero error for the $N = 2$ identity and a high-N same-AUC gap of 0.9988. The evidence is not trying to discredit AUC as a ranking diagnostic. It identifies its boundary: when N grows, the selected value depends on score-tail moments rather than only pairwise ordering. A reviewer can therefore accept this claim without believing any broad statement about all ranking metrics.

Claim: more imagined rollouts can reduce real utility. This claim is supported by severe mismatch, stuck/slip mismatch, anti-scorer, and randomized-dynamics artifacts. Severe mismatch produces gap growth 16.73, stuck/slip mismatch produces gap growth 16.39, and the anti-scorer moves from -14.10 at $N = 1$ to -26.57 at $N = 64$. These artifacts are intentionally negative. They show that the audit can diagnose harmful score tails rather than merely reporting favorable curves. The correct claim is conditional and operational: if the upper score tail is anti-aligned with real utility, increasing N will amplify the damage.

Claim: inference-value profiles can guide rollout-budget allocation. This claim is supported by the adaptive allocation artifact, where the reference allocation gain is 0.0767. The paper does not claim global optimality, adversarial robustness, or safe online control. The evidence is narrower and more useful: once an empirical profile has been estimated for a state or task family, marginal real-utility gains can be used to prioritize finite rollout budgets. The appendix keeps pilot noise, nonstationarity, and stale-profile failure modes visible so that the claim does not become a hidden theorem about arbitrary deployment.

Claim: the audit is not limited to a toy rollout pool. This claim is supported by benchmark breadth, not by any one benchmark alone. The v4 cache aggregates 8 benchmark rows, including state, visual, manipulation, kitchen-family, and LIBERO rollout-pool artifacts. RoboCasa coverage is particularly broad, with 97 tasks, 194 rollout pools, and 1552 heldout rollouts in the stratified artifact. The residual clean/cook artifact adds 35 more task slices. This is evidence that the audit transfers across simulator artifact families. It is not evidence of full RoboCasa-wide completion, hardware validation, or modern VLA policy performance.

Claim: the paper is auditable by a fresh session. This claim is supported by the source map, final PDF naming, v4 build script, v4 audit script, generated macros, and claim-status ledger. It is a practical submission claim: a new reviewer, author, or agent should be able to locate the final PDF, source folder, GitHub repository, result cache, and audit commands without reverse-engineering the repository history. This matters because a paper with many optional experiment branches can be scientifically strong but operationally confusing. V4 treats operational clarity as part of submission readiness.

T.2 Per-Benchmark Acceptance Criteria

The benchmark artifacts should be judged by criteria matched to the paper's claim, not by criteria borrowed from end-to-end robot-policy papers. A rollout-pool score-tail audit has three primary requirements. First, the finite identity must reproduce empirical best-of-N selection on the tested pool. Second, the scorer comparison must report whether the promoted score places real utility in the upper score tail. Third, unavailable modalities or runtime blockers must be labeled rather than hidden.

Fetch state and Fetch RGB. Fetch is acceptable in v4 if both state and visual variants show low exact-law error and positive learned-minus-random evidence under the rollout-pool protocol. The current artifacts satisfy this standard with exact-law MAEs 0.0126 and 0.0106, and CI lower bounds 0.441 and 0.348. The acceptance standard does not require the paper to claim a deployed visual policy. It requires the paper to show that the score-tail audit can be applied when the rollout evidence includes both compact state and RGB-derived features.

Meta-World and RoboSuite. Meta-World and RoboSuite are acceptable if they broaden manipulation coverage and preserve the distinction between exact accounting and score quality. The current artifacts report exact-law MAEs 0.0298 and 0.0024. If a learned score is weaker than a reward-like score in a slice, that should be reported as a diagnostic result rather than edited away. The point of the paper is not that every learned scorer is strong; the point is that the audit reveals whether the learned scorer controls the selected score tail.

ManiSkill. ManiSkill is acceptable only as state-mode evidence. The paper may report the state-mode exact-law MAE 0.0034 and discuss dense-score and learned-score behavior in that protocol. It may not imply visual or end-effector-control validation. The visual and EE-control blockers are part of the record. This is a useful stress case for claim discipline because a less careful paper could hide the blocked modalities behind a single "ManiSkill" label.

RoboCasa. RoboCasa is acceptable if breadth and incompleteness are reported together. The paper can claim stratified coverage over 97 tasks and 194 rollout pools, residual clean/cook coverage over 35 tasks, and catalog accounting over 396 registered task IDs. It must also state that this is not full RoboCasa-wide learned-WAM validation. The broad artifact is a major strength only because the scope boundary is explicit.

LIBERO. LIBERO is acceptable as a rollout-pool WAM-lite artifact and optional runtime branch. The paper can report 15 pools, correlation 0.353, and CI lower 0.265. It may also mention separate sparse-success smoke artifacts as runtime evidence. It must not describe those smokes as modern VLA benchmark performance. A strict reviewer should reward this boundary because it prevents a small optional artifact from being inflated into a headline result.

T.3 Stress Tests, Negative Controls, and Reviewer Attacks

The strongest version of this paper is not the version where every curve goes up. The strongest version is the one where the audit fails loudly when the score tail is bad. V4 therefore treats negative controls as first-class evidence. They are the difference between a paper that merely celebrates test-time sampling and a paper that tells a practitioner when test-time sampling should be stopped.

The first negative control is the anti-scorer. It deliberately makes high planner scores identify lower real utility. Under that condition, the finite law predicts that larger N will concentrate more probability on bad upper score groups. The observed movement from -14.10 to -26.57 is therefore not a surprising pathology. It is the expected behavior of max-score selection under a harmful score. A reviewer attacking the paper on this point should be answered directly: the anti-scorer is included so the paper does not pretend that more imagination is always useful.

The second negative control is imagined-real mismatch. A WAM-style planner may score futures according to imagined utility while the environment supplies a different real utility. High-N selection can then find rollouts that exploit the imagined score while losing real value. Severe mismatch and stuck/slip artifacts quantify that failure through large gap growth. These experiments are especially important because they are close to the robotics use case: planning errors often arise from model mismatch, not from an abstract ranking failure.

The third negative control is randomized or unsupported dynamics. If the rollout generator does not preserve the relevant transition structure, a score-tail identity can still account for selection from the pool, but the selected pool itself may be useless. This distinction is important enough to repeat: exact accounting is not score quality, and score quality is not world-model validity. V4 does not blur those layers. It uses exact-law errors to check accounting, scorer comparisons to check ranking, and mismatch artifacts to check whether imagined utility remains tied to real utility.

The fourth stress test is pilot staleness for adaptive budgeting. A marginal rollout-value profile is useful only if it estimates the current distribution. If the task distribution shifts, the allocator can spend compute on the wrong states. V4 therefore frames adaptive allocation as an audit-informed heuristic. A stronger future controller would need online uncertainty, drift detection, and a safe fallback when the estimated score tail becomes unreliable.

These stress tests make the manuscript harder to attack because they remove the easiest overclaim. A reviewer can still ask for hardware experiments or a new method that repairs bad score tails. That is fair. But the reviewer should not be able to say that the paper ignores harmful tails, hides score failure, or equates exact selection accounting with deployment safety. V4 says the opposite, repeatedly and with artifacts.

T.4 Expanded Area-Chair Decision Simulation

This section records the decision logic the paper should be able to survive. It is intentionally harsher than a normal author checklist because the project has a visible duplication risk from earlier "Best-of-N" drafts and because the evidence base contains many optional branches. The point is to make the submission robust before a real reviewer has to discover the weak spots.

Attack: this is just an order-statistics wrapper. The answer is that the theorem is simple but the paper is not selling the theorem as a universal novelty claim. The paper sells a WAM score-tail audit: measure the real utility of the rollouts that max-score planning will select, show how AUC fails beyond two samples, show harmful imagined-real tails, and apply the audit across simulator rollout pools. If the main text ever sounds like the contribution is only "we derived best-of-N," it should be revised. The v4 title, abstract, experiments, and appendix are designed to avoid that failure.

Attack: the evidence is not end-to-end robotics. The answer is to agree with the premise and defend the actual claim. The paper is a rollout-pool audit, not a hardware-control benchmark. It does not claim that a deployed controller solves Fetch, RoboCasa, or LIBERO. It claims that, given fixed candidate rollout pools and fixed scores, the real utility of max-score selection can be exactly audited and empirically stress-tested. That claim is meaningful for WAM deployment because rollout planners routinely spend test-time compute before knowing whether the upper score tail contains real utility.

Attack: the paper has too many artifacts and no center. The answer is the score-tail question: "When we ask a WAM planner to imagine more futures and pick the highest-scored one, what happens to real utility?" Every promoted artifact should answer one part of that question. Exact-law checks answer whether the accounting is correct. Same-AUC checks answer why tail moments matter. Mismatch and anti-scorer checks answer when high-N hurts. Benchmark artifacts answer whether the audit can be run outside a toy pool. Claim ledgers answer whether the manuscript is overstating those facts.

Attack: the paper is too cautious. The answer is that caution is the contribution's safety feature. A paper about score-tail audits should be conservative with claims because the whole point is to detect when the planner's attractive futures are not useful. The paper can be ambitious in evidence breadth while cautious in language. That combination is stronger than a broad claim that collapses under a single runtime blocker or missing hardware result.

Attack: the paper needs more experiments. The strongest response is to separate necessary from optional experiments. For v4, necessary experiments are exact-law validation, AUC-boundary validation, harmful-tail falsification, adaptive-budget evidence, and multi-suite rollout-pool checks. Those are present in committed artifacts. Optional experiments that would upgrade the claim tier include closed-loop hardware, full RoboCasa-wide validation, modern VLA candidate generators, and learned score repair. The paper should name those future upgrades without pretending they are already complete.

T.5 Large-Scale External Validation Protocol

If reviewers or future authors want to push the paper beyond v4, the next experiments should not be ad hoc. They should preserve the audit structure so that new evidence is comparable to the current claims. A large-scale external validation should start by fixing the candidate generator, scorer, task split, real utility definition, tie policy, and candidate-count grid. It should then write candidate-level records before aggregating any figure. The required columns are task ID, seed, state or episode ID, candidate ID, planner score, real utility, imagined utility if available, selected candidate under each N, and optional uncertainty or support features.

The first extension tier is larger heldout simulator coverage. RoboCasa is the obvious target because the current artifacts are broad but incomplete. A future run should define a heldout task-family split before training or scorer selection, generate rollout pools for every registered or selected task ID, and publish both completed and failed task IDs. It should keep the current coverage accounting rather than replacing it with a single success number. The audit should report exact-law MAE, learned-minus-random CI, oracle gap, negative-tail frequency, and task-family heterogeneity.

The second extension tier is modern VLA integration. A VLA validation should not reuse the current LIBERO smoke artifacts as a headline result. It should use a real candidate generator or policy, a fixed scorer or verifier, candidate-level rollout labels, and heldout task evaluation. The same score-tail law would then answer a sharper question: does the VLA's verifier place real task progress in the upper score tail when asked to choose among multiple proposed futures or actions?

The third extension tier is closed-loop deployment. Closed-loop validation should log each receding-horizon decision, the candidate pool available at that decision, the selected candidate, downstream real utility, and failure modes. Without those logs, an end-to-end success rate cannot be connected back to the score-tail mechanism. With those logs, the paper could test whether local inference-value profiles predict actual controller improvement and whether the planner should stop sampling in states with harmful score tails.

The fourth extension tier is score repair. The current paper audits the tail; it does not learn a universal repair. A future method could use the audit signals to calibrate scores, penalize unsupported imagined states, or allocate rollouts to states where the upper tail is reliable. Such a paper should be judged by improvement under heldout score-tail audits, not only by lower prediction error. The repair should improve real selected utility at larger N and reduce the negative-tail cases documented in v4.

This protocol is included because it prevents v4 from being both too small and too vague. The current paper is final as an audit paper, but the path to a larger method or deployment paper is explicit. Reviewers can therefore ask for a different claim tier without confusing that request with a flaw in the current theorem or artifact-backed audit.

T.6 How to Reject This Paper Correctly

A correct rejection would not say that the finite identity is false. The identity is exact under its stated assumptions. A correct rejection would say that exact accounting is too simple for the venue unless paired with a stronger method, stronger external validation, or stronger deployment study.

A correct rejection would not say that the paper claims high-N sampling is always good. The paper explicitly shows that anti-aligned score tails can make high-N selection worse. A correct rejection would say that the harmful-tail cases are still controlled or simulator-backed and need real deployment evidence.

A correct rejection would not say that the paper proves modern VLA performance. The paper does not claim that. A correct rejection would say that the optional LIBERO and RoboCasa artifacts are not enough for a venue that expects end-to-end policy benchmarking.

A correct rejection would not say that the paper hides limitations. The paper states no hardware validation, no full RoboCasa-wide validation, no ManiSkill visual/EE-control validation, and no universal WAM training law. A correct rejection would say that these limitations are too large for the target venue.

Naming correct rejection paths helps v4 avoid dishonest defenses. It also clarifies what a future v4 would need: hardware or high-fidelity closed-loop evaluation, a stronger learned scorer, broader exact rollout-pool labels under heldout task splits, or a method that actively repairs harmful score tails.

T.7 HOW TO ACCEPT THIS PAPER CORRECTLY

A correct acceptance would view the paper as a deployment-audit contribution. The paper gives an exact score-tail accounting identity for fixed rollout/scorer stacks, shows where AUC stops being sufficient, demonstrates that imagined-real mismatch can make additional rollouts harmful, and checks the audit across a broad simulator artifact base. The contribution is not a new dynamics model or a universal robot policy.

A correct acceptance would also value the negative evidence. The anti-scorer and mismatch artifacts are not embarrassing edge cases; they are why the audit is needed. A planner that spends test-time compute without measuring real utility in the upper score tail can become worse as N grows. This is a practical warning for WAM-style deployment.

Finally, a correct acceptance would value the claim discipline. The repository contains many optional branches, but v4 does not inflate them. It reports RoboCasa coverage counts, LIBERO scope boundaries, ManiSkill blockers, and claim-status checks. This makes the paper easier to trust because the authors are not asking reviewers to infer the boundary from scattered artifacts.

T.8 FUTURE WORK THAT WOULD CHANGE THE CLAIM TIER

The most valuable future work would add online evaluation where the planner's selected action is executed in a closed-loop environment and delayed real utility is logged. This would move part of the evidence from rollout-pool audit toward controller evaluation. The key would be to retain candidate-level logs: scores, selected candidates, real outcomes, uncertainty, and support metrics.

A second future direction is score repair. V4 audits score tails and includes pilot-calibrated repair artifacts, but it does not propose a general learned repair method. A stronger method paper could use the audit to train or calibrate scores that avoid harmful high-N tails, then evaluate those repairs under heldout task families.

A third future direction is modern VLA integration. This would require a real VLA candidate generator or policy, a scorer or verifier, candidate-level rollout labels, and task-level heldout evaluation. It should not simply reuse the v4 LIBERO smoke artifacts. It should create a new evidence tier with its own claim boundaries.

A fourth future direction is full RoboCasa-wide coverage. The current coverage is broad but incomplete. Full coverage would require rollout-pool artifacts for all registered tasks or a principled stratified claim explaining which task families are covered and why. V4's coverage accounting is a starting point, not the finish line.

T.9 FRESH-SESSION WORKFLOW

A fresh agent should follow a short deterministic workflow. First, read `PAPER_SOURCE_MAP.md` on the Desktop and locate `best-of-n-wam-v4.pdf`, this local folder, and the GitHub repository. Second, run `gitstatus--short--branch` and verify that the working tree is clean. Third, inspect `paper/final/best-of-n-wam-v4.pdf` rather than the unversioned build product. Fourth, run `pythonscripts/run_v4_claim_audit.py` to check the page count, hashes, source map, and evidence summary. Fifth, only then inspect or modify the manuscript.

This workflow matters because the repository has many reports and historical artifacts. Without a final source map and v4 audit, a new session could easily open an old PDF, treat a smoke artifact as final evidence, or confuse optional runtime probes with promoted claims. The v4 archive contract is meant to remove that ambiguity.

T.10 ANONYMITY, REPRODUCIBILITY, ETHICS, AND LLM DISCLOSURE

Anonymity and supplement plan. The submission source avoids author names, institution names, non-anonymous repository links, and private machine paths. The intended supplement is an anonymous code archive containing theorem implementations, scripts, result tables, figure-generation scripts, and a manifest mapping claims to artifacts. Any public repository link should be anonymized before upload.

Reproducibility statement. The theorem assumptions are stated in the main paper and proved in Appendix A. Experiments are reported as artifact-backed checks, with result tables and finite-identity summaries referenced in Appendix B. The current local artifact registry reports verified coverage for canonical experiment families, scripts, tables, figures, and claim checks. Before upload, authors should run the claim-status and result-consistency scripts on the anonymous artifact bundle and include the exact command log in the supplement.

Ethics statement. The work is theoretical and simulator-based. It does not report human-subject data or real-world robot deployment. A practical risk is that high-N planning can amplify a misaligned scorer or unsafe model error; the paper explicitly frames this as a diagnostic risk and recommends treating high-score-tail audits as evidence, not as safety certification.

LLM usage disclosure. This draft was materially assisted by a large language model for ideation, prose drafting, code navigation, and build orchestration. The authors remain responsible for checking every theorem, citation, artifact, and empirical claim before submission.

T.11 CLAIM BOUNDARIES

The paper should preserve the following boundaries in any future revision.

- The exact claims are conditional on a fixed rollout generator, fixed scorer, fixed state distribution or empirical pool, sampling with replacement, and stated tie-breaking.
- The continuous laws are no-tie population identities and should not be substituted for finite tied empirical pools.
- The $N = 2$ AUC identity is exact for $N = 2$ only. For larger N, AUC alone is insufficient.
- Adaptive allocation is an inference-budgeting diagnostic supported by tested artifacts, not a proof of a global controller guarantee.
- Current evidence is simulator and benchmark rollout-pool evidence, not real-world robot validation.